

Design and Evaluate Different Prompt Types to Control Agent Reasoning, Autonomy, and Output Quality

Objective

Evaluate how different prompting techniques influence reasoning, autonomy, and output quality of AI agents.

Prompt Types

Prompt Type	Description	Example
Zero-shot	Task only, no examples.	Summarize this article in 100 words.
Few-shot	Provide examples before the task.	Example summaries then a new article.
Chain-of-Thought	Encourage step-by-step reasoning.	Solve step by step.
Role-based	Assign an expert role.	You are an experienced software architect.
Instruction-based	Detailed instructions and constraints.	Generate Python code with comments.
Self-Reflection	Review and improve own answer.	Review your solution and improve it.
ReAct	Reason while taking actions/tools.	Think, search if needed, then answer.
Tree-of-Thought	Explore multiple solution paths.	Generate three solutions and compare.

Evaluation Criteria

Accuracy: Correctness of the response. **Completeness:** Covers all requested information.

Reasoning: Logical explanation. **Autonomy:** Ability to make decisions independently.

Hallucination: Lower is better. **Clarity:** Easy to understand. **Response Time:** Lower is better.

Sample Evaluation Results

Prompt	Reasoning	Autonomy	Output Quality	Overall
Zero-shot	2/5	2/5	3/5	2.3
Few-shot	3/5	3/5	4/5	3.3
Role-based	4/5	3/5	4/5	3.7
Instruction-based	4/5	4/5	5/5	4.3
Chain-of-Thought	5/5	4/5	5/5	4.7
Self-Reflection	5/5	5/5	5/5	5.0
ReAct	5/5	5/5	5/5	5.0
Tree-of-Thought	5/5	5/5	5/5	5.0

Conclusion

Chain-of-Thought, ReAct, Tree-of-Thought, and Self-Reflection provide the strongest reasoning and autonomy. Instruction-based prompting offers excellent control over format and constraints. Zero-shot prompting is suitable for simple tasks, while complex agentic workflows benefit from advanced prompting strategies.